

Hee-Seung Moon

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CURRENT POSITION

Chung-Ang University, Republic of Korea
Assistant Professor

September 2024 – Now
School of Computer Science and Engineering

- Leading *Computational Human–AI Interaction Laboratory (CHAI lab)*
- Research Interests: Human–Computer Interaction, Computational Interaction, User Behavior Modeling

EDUCATION

Yonsei University, Republic of Korea
Ph.D., School of Integrated Technology, College of Engineering
· Thesis Title: Adaptation of Deep User Behavior Model for Personalized Interaction

March 2015 – August 2022
Advisor: Jiwon Seo & Byungjoo Lee

Yonsei University, Republic of Korea
B.S., School of Integrated Technology, College of Engineering

March 2012 – February 2015

EXPERIENCE

Aalto University, Finland
Postdoctoral Researcher

September 2022 – August 2024
Computational Behavior Lab (Advisor: Antti Oulasvirta)

ETH Zürich, Switzerland
Visiting Researcher

May – July 2024
Sensing, Interaction and Perception Lab (Advisor: Christian Holz)

Aalto University, Finland
Visiting Researcher

March – May 2022
User Interfaces Research Group (Advisor: Antti Oulasvirta)

Naver AI Lab, Republic of Korea
Research Intern

April – October 2021
Mentor: Minsuk Chang

Yonsei University, Republic of Korea
Doctoral Student Researcher

March 2015 – August 2022
Intelligent Unmanned Systems Lab (Advisor: Jiwon Seo)
Esports & High-Performance HCI Lab (Co-advisor: Byungjoo Lee)

PUBLICATIONS

Point & Grasp: Flexible Selection of Out-of-Reach Objects Through Probabilistic Cue Integration

X. Luo, **H.-S. Moon***, C. Holz, and A. Oulasvirta (*Corresponding Author)

Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI 2026)

Efficient Human-in-the-Loop Optimization via Priors Learned from User Models

Y.-C. Liao, J. Belo, **H.-S. Moon**, J. Steimle, and A. M. Feit

Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI 2026)

Modeling User Performance in Multi-Lane Moving-Target Acquisition

J. Kim, J. Kim, J.-S. Yoon, **H.-S. Moon**, S. Kim, and B. Lee

Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI 2025)

Modeling Visually-Guided Aim-and-Shoot Behavior in First-Person Shooters

J.-S. Yoon, **H.-S. Moon**, B. Boudaoud, J. Spjut, I. Frosio, B. Lee*, and J. Kim* (*Co-corresponding Authors)

International Journal of Human-Computer Studies, vol. 199, 2025

WigglyEyes: Inferring Eye Movements from Keypress Data

Y. Zhu, D. Shi, **H.-S. Moon**, and A. Oulasvirta

Proceedings of the 2025 ACM International Symposium on Wearable Computers (ISWC 2025)

Real-time 3D Target Inference via Biomechanical Simulation

H.-S. Moon, Y.-C. Liao, C. Li, B. Lee, and A. Oulasvirta

Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI 2024)

****Best Paper Honorable Mention (Top 5%)****

Amortized Inference with User Simulations

H.-S. Moon, A. Oulasvirta, and B. Lee

Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI 2023)

Speeding up Inference with User Simulators through Policy Modulation

H.-S. Moon, S. Do, W. Kim, J. Seo*, M. Chang*, and B. Lee* (*Co-corresponding Authors)

Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI 2022)

Fast User Adaptation for Human Motion Prediction in Physical Human–Robot Interaction

H.-S. Moon and J. Seo

IEEE Robotics and Automation Letters (RA-L), vol. 7, no. 1, 2022

Sample-Efficient Training of Robotic Guide Using Human Path Prediction Network

H.-S. Moon and J. Seo

IEEE Access, vol. 10, 2022

Optimal Action-based or User Prediction-based Haptic Guidance: Can You Do Even Better?

H.-S. Moon and J. Seo

Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI 2021)

Prediction of Human Trajectory Following a Haptic Robotic Guide Using Recurrent Neural Networks

H.-S. Moon and J. Seo

2019 IEEE World Haptics Conference (WHC)

Effect of Redundant Haptic Information on Task Performance during Visuo-Tactile Task Interruption and Recovery

H.-S. Moon, J. Baek, and J. Seo

Frontiers in Psychology, vol. 7, art. 1924, 2016

GRANTS AND FELLOWSHIPS

Outstanding Young Scientist Grant <i>National Research Foundation of Korea</i> · Proposal Title: Biomechanical Simulation Intelligence for Human–AI Cooperation	2025 - 2028
International Postdoc Fellowship <i>National Research Foundation of Korea</i> · Proposal Title: Inferring User Input Intention in VR based on Biomechanical Simulation	2023 - 2024
Graduate Fellowship <i>ICT Consilience Creative Program, Ministry of Science and ICT, Republic of Korea</i>	2015 – 2019
Undergraduate Fellowship <i>ICT Consilience Creative Program, Ministry of Science and ICT, Republic of Korea</i>	2012 – 2015

AWARDS AND HONORS

CHI 2024 Best Paper Honorable Mention · Paper: Real-time 3D Target Inference via Biomechanical Simulation	2024
Excellent Academic Paper Award <i>Yonsei University, Republic of Korea</i> · Paper: Sample-Efficient Training of Robotic Guide Using Human Path Prediction Network (<i>CHI 2022</i>)	2022
Minister Award Ministry of Science and ICT, Republic of Korea	2014

ACADEMIC SERVICE

Associate Chair (AC) · ACM CHI 2026 (<i>Computational Interaction</i> Subcommittee)
Reviewing · 30+ reviews on top-tier HCI venues (e.g., CHI, UIST) · 10+ <i>Special Recognitions for Outstanding Reviews</i>

TEACHING

Instructor · Artificial Intelligence (CSE 17182, Chung-Ang Univ.) · Human–Computer Interaction (CSE 17130, Chung-Ang Univ.) · Big Data Reinforcement Learning (CSE 58374, Chung-Ang Univ.) · Compilers (CSE 52321, Chung-Ang Univ.) · Automata and Formal Languages (CSE 40458, Chung-Ang Univ.)	Fall 2025, Spring 2026 Spring 2025, 2026 Fall 2024, 2025 Spring 2025 Fall 2024
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